DATA SCIENCE CASE STUDY

Why customers leave, and who to keep first

A churn analysis on ~7,000 telecom customers: the few dynamics that drive attrition, and a concrete retention plan.

Presented to the business owner of the retention program

CONTEXT & THE QUESTION

A quarter of customers leave, and the operator wants to know who and why

The client, a telecom operator, is losing customers and wants to protect recurring revenue. Two questions were set at kick-off:

1

What makes a customer likely to cancel?

Identify the characteristics that drive attrition.

2

Who is going to cancel?

Predict churn to prioritise retention effort.

26.5%

of customers churned in the sample

Not a niche problem, but a highly concentrated one.

THE DATA

One row per customer: who they are, their contract, their services

7,043
customers (one row each)

19
descriptive features

1
target: churned yes / no

Snapshot
no history / no timestamps

Demographics

gender, senior citizen, partner, dependents

Contract

tenure, contract type, billing, payment method

Charges

monthly charges, total charges

Services

phone, internet, security, backup, streaming...

THE DATA · DICTIONARY

Every field in the dataset, at a glance

Field	Description
Unnamed: 0	Row index
customerID	Unique client identifier
gender	Sex
SeniorCitizen	Whether the client is a senior
Partner	Whether the client has a partner
Dependents	Has dependents (e.g. children)
tenure	Months as a customer (contract age)
PhoneService	Has phone service
MultipleLines	Has multiple phone lines
InternetService	Internet type: DSL, fibre, none
OnlineSecurity	Subscribed to security (firewall)

Field	Description
OnlineBackup	Subscribed to backup service
DeviceProtection	Subscribed to device protection
TechSupport	Subscribed to tech support
StreamingTV	Uses TV streaming
StreamingMovies	Uses movie streaming
Contract	Contract type: monthly, 1-yr, 2-yr
PaperlessBilling	Paperless vs paper billing
PaymentMethod	Payment method
MonthlyCharges	Monthly subscription price
TotalCharges	Total paid since contract start
Churn	Target: did the customer leave

Unnamed: 0 and **customerID** *are identifiers, dropped before modelling.*

Three cleaning decisions, each tied to what a missing value means

1

Total charges missing for 2,122 rows

All were brand-new customers (tenure = 0). Rebuilt as tenure × monthly charge rather than dropped. This keeps the new-customer segment, which turns out to matter most.

2

Services blank when there's no internet

1,500+ blanks were simply customers without internet. Relabelled "No internet service" so the model reads absence correctly, not as unknown.

3

Online backup unknown for 4,400 customers

These have internet but a genuinely missing value. Kept as "Unknown" instead of forcing a No. A wrong guess here would bias the whole variable.

Principle: a missing value is not neutral; its meaning depends on why it's missing.

APPROACH

From raw signals to a decision, in three steps

STEP 1

Describe

Which customers churn more? Simple, transparent comparisons across every attribute.

EDA

STEP 2

Explain

Separate real drivers from coincidences by controlling for confounders (e.g. new customers ≠ contract effect).

Adjusted model

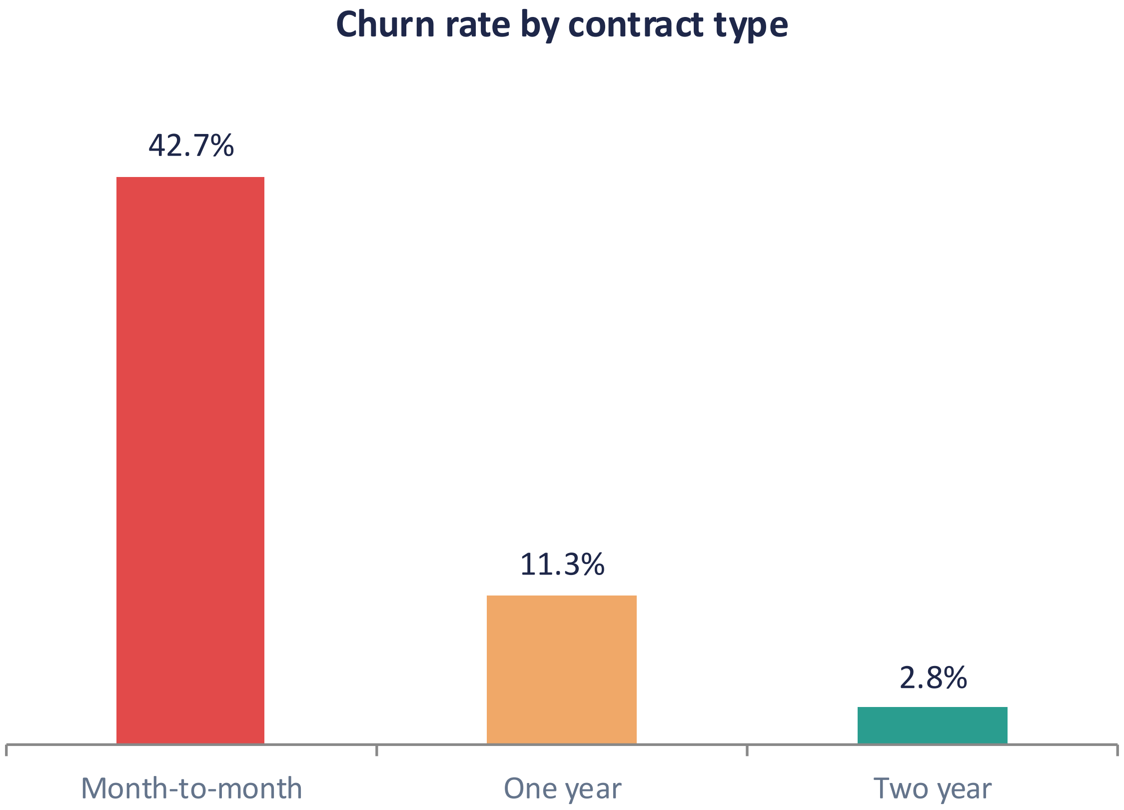
STEP 3

Act

Rank customers by risk so a limited retention budget hits the right people first.

Targeting

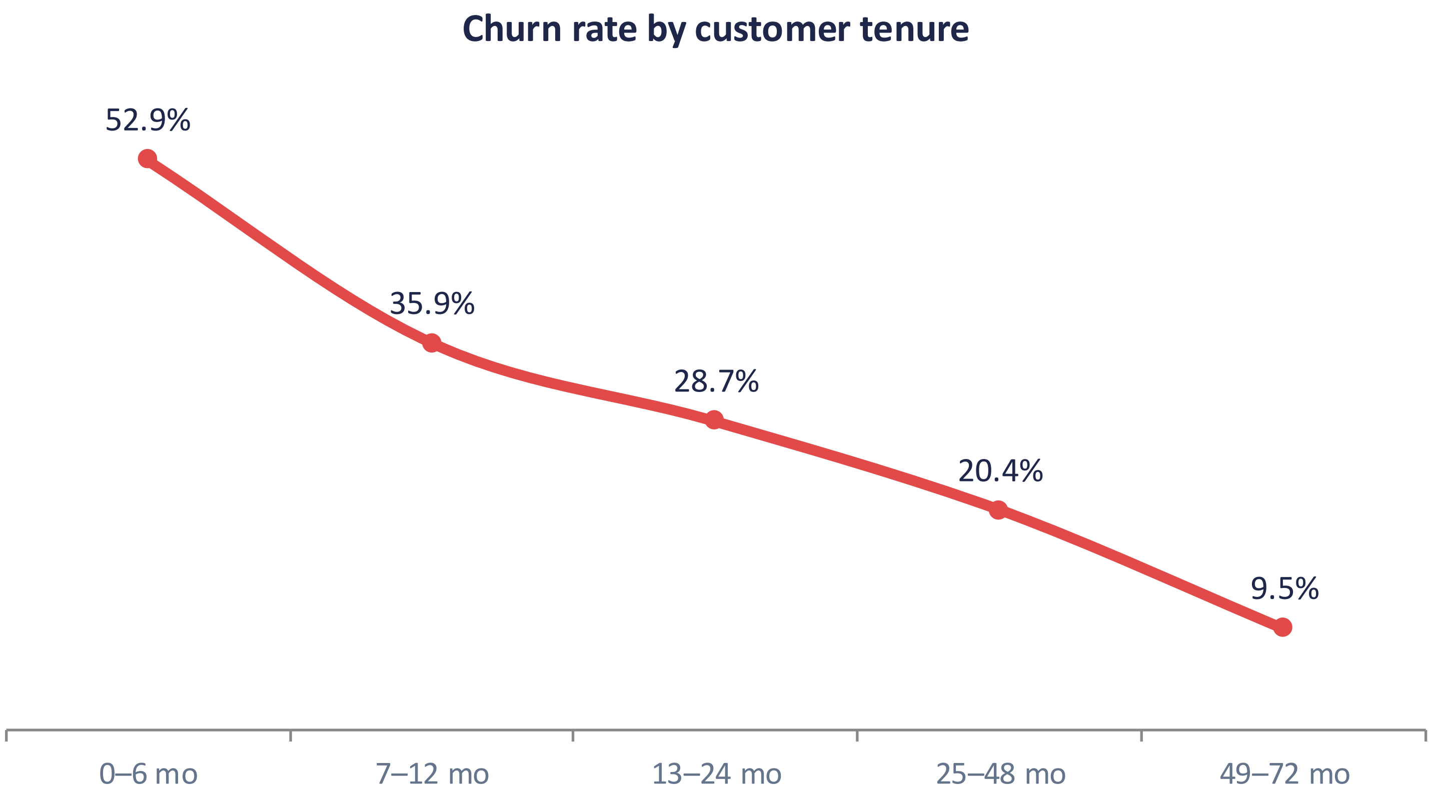
The same names keep coming up: contract, fibre, payment, being new



Other strong signals

Fibre-optic internet	**41.9%**	vs 19% on DSL
Electronic-check payment	**45.3%**	vs ~16% on autopay
Senior citizens	**41.7%**	vs 23.6% otherwise
First 6 months	**52.9%**	the danger zone

Churn is front-loaded: it happens in the first months, then fades



~42%

of all churn happens in a customer's first 6 months

Retention won or lost early: this is an onboarding problem before it's a loyalty problem.

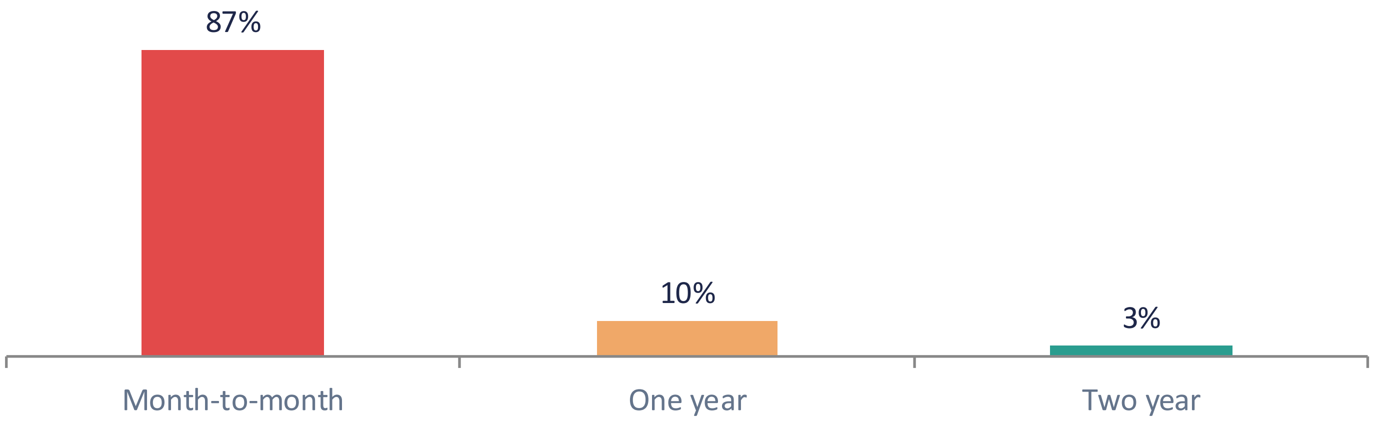
Counting customers understates it: churners are the higher-value ones

30.5%
of monthly recurring revenue is at risk

€74 vs €61
avg. monthly bill: churners vs those who stay

87%
of that revenue-at-risk sits in month-to-month plans

Where the revenue-at-risk is concentrated



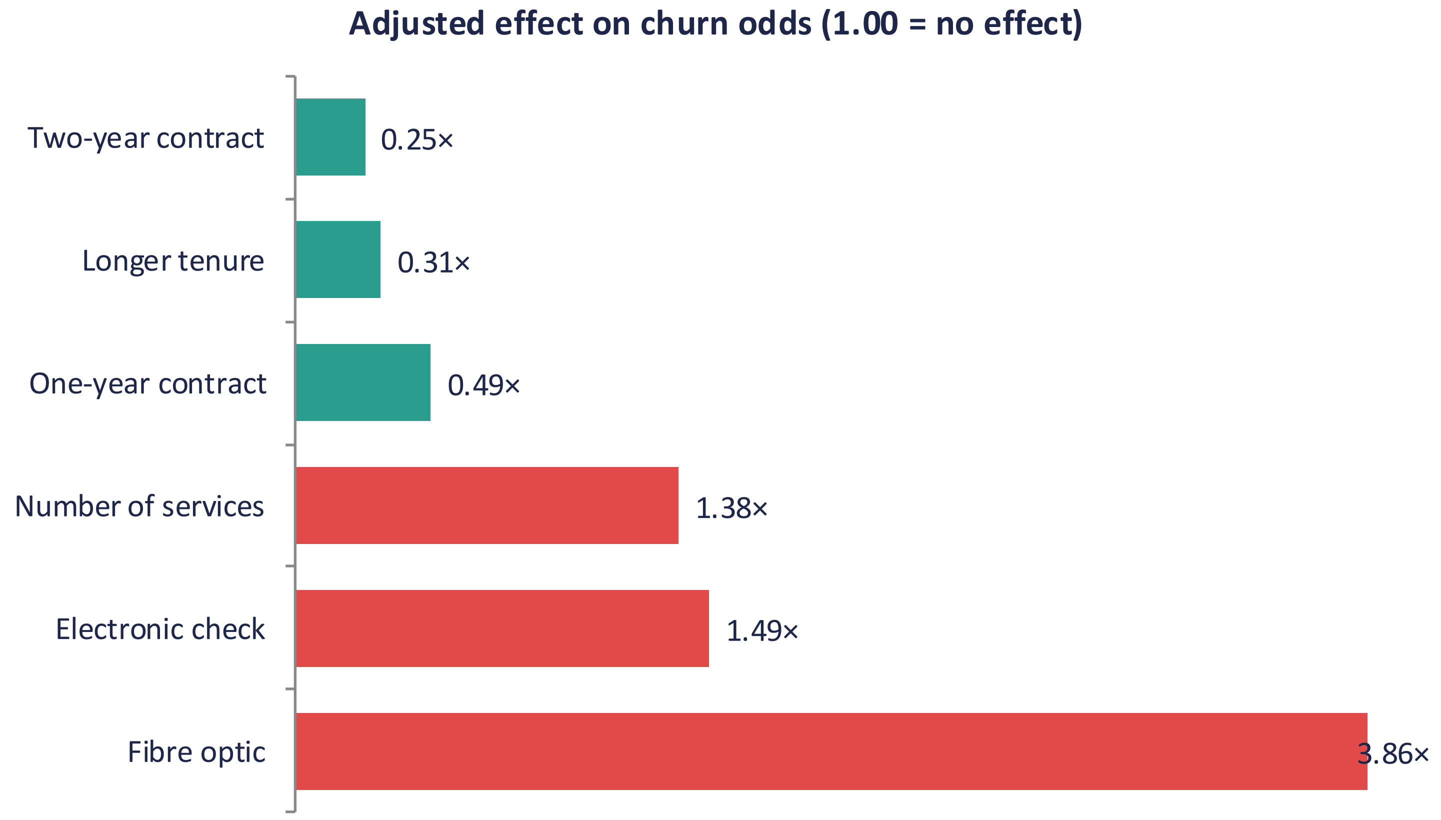
The sharpest pocket

Month-to-month + fibre + first 6 months

619 customers · 74% churn rate
A small, identifiable group doing outsized damage.

Once we control for everything else, four levers really move churn

These effects come from a logistic regression that weighs every factor at once; it's the same model used on the next slide to rank customers by risk.



Reading it

Above 1 raises churn; below 1 lowers it, each holding all else equal.

Contract is the biggest lever

A two-year contract cuts churn odds ~75%.

Fibre is a real red flag

Still ~3.9× churn even after price & tenure: a product/experience issue, not just new customers.

Client's hypothesis: nuanced

"More services" does raise churn, but the raw link is non-linear, not a clean rule.

A simple model ranks customers so limited effort reaches the right ones

0.84

model AUC (vs 0.50 baseline)

~50%

of churners reached by contacting the top 20%

2.5×

more precise than contacting at random

“Top 10%” = the 10% of customers the model ranks as most likely to churn (then 20%, 30%).

In practice

A logistic regression. What matters isn't a verdict on each customer; it's the ranking, so a fixed budget reaches the most churners.

Four moves, ordered by impact and ease

1 Own the first 6 months

Biggest lever

Structured onboarding for new customers: check-ins, early-value nudges. This is where ~42% of churn is decided.

2 Convert month-to-month to annual

High impact

Targeted incentives to migrate, starting with fibre customers. Contract is the strongest controllable driver.

3 Investigate the fibre experience

Deep dive

Fibre churns ~3.9× even after price. Dig into quality, complaints, install experience before it costs more.

4 Deploy the risk list

Quick win

Feed the top-risk quintile to retention each month: half the churn, a fraction of the effort.

What this analysis can't yet claim, and how to close the gap

Limitations

- **Snapshot only**
  No dates or event history, so we see who left, not when or why.
- **Correlation, not proof**
  Effects are associations; an action isn't guaranteed to cause retention.
- **Fibre stays ambiguous**
  Price and product quality can't be fully separated with these fields.

Next steps

- **Test, don't just target**
  A/B the onboarding & migration offers to measure real causal lift.
- **Add behavioural data**
  Usage and support tickets to explain the fibre effect.
- **Model time-to-churn**
  Survival analysis to know when to intervene, not just who.
- **Refresh on live cohorts**
  Re-fit on recent customers and monitor drift.

IN ONE MINUTE

Churn is concentrated, so retention can be too

The question

Understand why ~26.5% of customers leave, and predict who will.

What we found

Risk clusters in month-to-month, fibre, and the first 6 months, which are also the higher-value customers (30% of revenue).

The lever

Contract length and early onboarding move churn most, once confounders are stripped out.

What to do

Onboard early, migrate to annual plans, investigate fibre, and work a model-ranked call list: top 20% ≈ half the churn.